
Call Laura: A Deterministic, Attribution-Anchored Multi-Agent Framework for Document Review

Paper 3 of 3 — building on Human–AI Interaction Emergent Co-Evolution (Serna Gaviria & Kepp, 2026, “Paper 1”) and Jarvis (Serna Gaviria & Kepp, 2026, “Paper 2”)

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Abstract

Paper 1 introduced two named instruments — the 8-Layer Cognitive Model and the User Integrity Protocol (UIP) — as ways of reading structure and integrity into a document or dialogue, and Paper 2 showed how a third instrument, the Continuous Co-Evolution Tracker, could be operationalized as mechanically computed software rather than left as self-reported narration. This paper documents a fourth operationalization, applied to a different problem: reviewing an arbitrary plan or document, once, and returning findings a reader can actually check.

Existing document-review tooling built on language models suffers from one of two failure modes: either a single model call returns a blended verdict with no traceable link from a claim back to the evidence that produced it, or a multi-agent ensemble of model calls compounds the same problem across several unreadable reasoning chains, with no way for a user to tell whether any given finding reflects a named researcher’s methodology or the tool’s own invention. We describe **call-laura**, a small, deterministic multi-agent framework — an “agent team,” each agent a pure function of the submitted text — that applies Paper 1’s 8-Layer Model and UIP faithfully, adds two further lenses of the project’s own, and attaches to *every single finding* a mandatory, type-enforced provenance tag stating exactly whose methodology produced it: the primary author’s own published framework, this project’s own operationalization of a concept she names but does not herself specify an algorithm for, or an addition with no connection to her work at all. We further describe its extension, a fifteen-agent domain-specialist orchestration layer (“the SWAT team”) that runs the same evidence-anchored, attribution-tagged discipline across a much wider set of concerns — security,

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legal and regulatory exposure, finance, operations, accessibility, and more — synthesizing convergent findings across agents into a single prioritized review rather than fifteen disconnected opinions.

The central claim of this paper is narrower than it might sound: not that keyword and pattern matching understands a document better than a language model does — it does not, and we say so throughout — but that a review tool bearing a named researcher’s identity has no integrity unless a reader can verify, per finding, whether a given claim is really her work or this project’s own judgment call. Determinism is not offered here as a claim of superior insight; it is offered as the price of that verifiability being possible at all.

Keywords. *deterministic multi-agent review, attribution-anchored findings, evidence-anchored document analysis, provenance tagging, honest partial failure, Model Context Protocol, 8-Layer Model operationalization.*

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1 Introduction

A recurring theme across this series is the gap between a theoretical instrument and its operational form. Paper 1 named the 8-Layer Model and the UIP as ways of reading a document or dialogue; it did not, and — for a single-subject pilot study — could not, specify an algorithm for applying either one mechanically to arbitrary new text. Paper 2 closed an analogous gap for the Continuous Co-Evolution Tracker. This paper closes it for document review: given a plan, a proposal, or any other piece of prose, how should a piece of software actually decide which of its layers are present, which claims are unsupported, and where its structure has broken down

— and how should it disclose, per finding, whether that decision reflects the named researcher’s own framework or the software’s own operational choice?

The problem this paper is actually solving is one of trust architecture, not natural-language understanding. Two failure modes dominate existing document-review tooling built on language models. An **opaque single-model review** returns one blended score or one paragraph of free-text verdict, with no way to trace any individual claim back to the span of the document it is reacting to, and no way to tell whether the model’s judgment reflects any particular named methodology at all. An **opaque multi-agent ensemble** — several model calls dressed up as “agents” — inherits the identical problem multiplied: each agent’s internal reasoning is still an unreadable model call, and there is still no per-finding record of whose methodology, if anyone’s, produced a given result. Both failure modes share a root cause: nothing in the pipeline is required to say, for any single output, where it came from.

`call-laura` is built around a specific answer to that: make the provenance tag mandatory at the type level, not a documentation convention. Every finding this framework produces carries a `source` field that is exactly one of three values, and the type system will not compile a finding that omits it. This paper describes the resulting architecture, its four founding lenses, its evidence-anchoring discipline, its honest-partial-failure behavior, its dual delivery as a local library and a hosted service, and its extension into a fifteen-agent specialist orchestration layer — always keeping the same question in view: for any given finding, can a reader tell whose judgment it actually represents?

2 Grounding in Papers 1 and 2

Two constructs are inherited directly from Paper 1 and are described here only as far as needed to explain how they are operationalized; see Paper 1 §1 and its Appendix A for their full theoretical treatment.

- **The 8-Layer Cognitive Model** (facts, analysis, patterns, hypotheses, symbols, action, counterarguments, blind spot) — in Paper 1, a structural analysis procedure applied by the primary author, by hand, to her own dialogue corpus. This paper’s `eight_layer` lens (§4.1) is the first operationalization of that taxonomy applied to *arbitrary* documents rather than to the author’s own interaction history.
- **The User Integrity Protocol (UIP)** — in Paper 1, a normative framework for semantic and ethical stability (no reinterpretation of explicit negations, no fabrication, triple verification, auditable outputs). This paper’s `uip_check` lens (§4.1) operationalizes four of its rules as pattern checks: stated constraints, unsupported absolute claims, unverified “done” claims, and un-auditable conclusions.

Paper 2 established a separate but related discipline: that an instrument built to measure a theoretical construct should be honest, in its own documentation, about exactly which parts of its behavior come from the cited theory and which are the implementers’ own choice. This paper applies that same discipline to a review framework rather than a measurement tracker, and — because a review tool that bears a named person’s identity carries a sharper version of the same honesty obligation — takes the additional step of enforcing the distinction in the software’s type system, not only in its documentation.

3 System Overview

`call-laura` is a Rust workspace with three tiers:

1. **call-laura-core** — the deterministic lens logic: four independent, pure-function review agents (`eight_layer`, `uip_check`, `resonance`, `ecocentric`), each mapping input text to a list of evidence-anchored findings, assembled by a synchronous dispatcher with no I/O. Open-core, published on crates.io.
2. **laura-mcp** / **laura-api** — the same core delivered two ways: a stdio Model Context Protocol server any MCP-connected agent can install with zero configuration, and a hosted HTTP service exposing the identical review as a keyed REST endpoint and a keyless MCP-over-HTTP endpoint.
3. **laura-team** — a fifteen-agent domain-specialist extension (§5) running the same discipline across a far wider set of review concerns, forming the platform’s paid analysis module.

All three tiers share one non-negotiable property: no lens, at any tier, ever calls a language model or makes a network request to produce a finding. Every finding is constructed from a plain, source-readable pattern match against the submitted text.

4 Detailed Architecture

4.1 The agent team and its four founding lenses

A document is submitted once; a dispatcher runs each requested lens (by default, all four) as an independent pure function of the text and assembles the results into one review with a summary of flag/note counts and any lens that failed to run. Each lens carries a distinct methodological source:

Lens	What it checks	Mechanism	Source
<code>eight_layer</code>	Classifies each document segment against the 8-Layer Model; flags entirely-absent layers	Keyword triggers per layer, applied per segment	<code>laura-8layer-2025</code> — category <i>names</i> are the primary author’s; the classification method is this project’s operationalization
<code>uip_check</code>	Four UIP rules: stated constraints, unsupported absolute claims, unverified “done” claims, un-auditable conclusions	Sentence-level pattern matching	<code>laura-uip-2025</code>
<code>resonance</code>	Low lexical overlap between sections that should plausibly agree (e.g. goals vs. success criteria)	Local term-frequency cosine similarity	<code>rfi-irfos-operationalization</code> — same general idea as Paper 2’s CCET, applied to a different question with a cruder mechanism — explicitly <i>not</i> CCET
<code>ecocentric</code>	Missing environmental, downstream, long-term, or systemic consideration	Keyword-category presence check	<code>rfi-irfos-addition</code> — no connection to the primary author’s research

Table 1: The four founding lenses of `call-laura-core` and their attribution source.

4.2 Determinism and inspectability

Each lens is keyword or pattern matching over the input text, with the exact trigger lists readable directly in source. The same document always yields the same review: no language model, no network call, no API key, at analysis time. This was not the original design — an earlier version called an NVIDIA-hosted model once per lens — but a deliberate rewrite after that provider’s account hit a persistent entitlement error unrelated to the tool’s own logic. The rewrite turned out to be a genuine improvement rather than a workaround: zero marginal cost, zero external

dependency, and every decision now inspectable in source rather than living inside a model call nobody can read.

4.3 Evidence anchoring

Every finding carries a **claim**, an **evidence** field that is a verbatim span quoted from the submitted input, and a **severity**. The **uip_check** and **ecocentric** lenses in particular construct findings *only* from spans actually present in the text — there is no separate “generate a claim, then go find supporting evidence” step that could produce a claim disconnected from what it cites. The finding *is* the matched sentence, which closes off an entire class of hallucination risk by construction rather than by review.

4.4 Mandatory per-finding attribution — the central mechanism

Every **LensResult** a lens produces carries a **source** field of a closed, three-valued type: the primary author’s own published framework, this project’s operationalization of a concept she names but does not herself algorithmically specify, or an addition with no connection to her work. This is not a convention documented in a README that a future contributor could forget to follow — it is required by the data model itself, so a lens that omitted it would not compile. Alongside **source**, every lens also carries a plain-language **attribution_note** stating what that lens’s mechanism specifically cannot do (miss things phrased without trigger words, false-positive on a trigger word used in an unrelated sense, no genuine semantic understanding) — an explicit statement of the trade being made, rather than a confident-sounding output that implies more than pattern matching can deliver.

This mechanism is this paper’s actual novel contribution, more than any individual lens. Prior multi-agent review tooling can report *that* an agent flagged something; this framework additionally guarantees, at the type level, that a reader can determine *whose methodology* is responsible for every single flag, every time, without exception.

4.5 Honest partial failure

A lens with nothing classifiable in its input — the **resonance** lens against a single-section document, for instance — reports its own **error** field rather than a fabricated result, and the assembled review’s summary is explicitly marked partial rather than presenting a clean bill of health. Because each lens runs independently, one lens’s failure to classify never silently degrades or contaminates another lens’s output.

4.6 Dual delivery and the paid module

The identical deterministic core is delivered three ways: as a Rust library any Rust project can embed directly; as **laura-mcp**, a stdio Model Context Protocol server installable with a single command and zero environment configuration (there is nothing to configure — the core has no external dependency at review time); and as **laura-api**, a Fly.io-hosted service exposing the same review over **POST /review** (REST, key-gated) and **POST /mcp** (MCP-over-HTTP, keyless, rate-limited for abuse hygiene rather than cost — there is no per-request external cost anymore). The hosted service is the mechanism through which this framework forms the Emergent Interaction Lab platform’s paid analysis module: reviewing a document is a service offered to clients, priced and delivered through the platform’s existing commercial infrastructure, with the free, deterministic four-lens core remaining openly licensed regardless of who is paying for the hosted convenience of using it.

5 Extension: The Fifteen-Agent Specialist Orchestration Layer

The four founding lenses cover a narrow, specific set of structural and integrity concerns. Real-world document review routinely needs to cover far more ground at once — security posture, legal and regulatory exposure, financial claims, operational risk, hiring language, accessibility, and more — and a single generalist pass over all of those concerns at once tends either to miss whole categories or to blur them into one vague, unaccountable verdict.

`laura-team` runs fifteen domain-specialist agents over the same submitted text in parallel — OSINT and exposure, security, legal and compliance, finance, operations, strategy, UX and accessibility, data privacy, ethics, threat modeling, reliability, hiring and people, brand and communications, research methodology, and product — each one a deterministic pure function of the input, in the exact same sense as the four founding lenses: no network call, no model call, evidence anchored to a verbatim span, and a `Source` tag (here, uniformly `rfi-irfos-addition`, since none of the fifteen specialist lenses derive from the primary author’s published research). What makes this more than fifteen independent tools bundled together is the orchestration layer above them: a synthesis step identifies **cross-cutting themes** — regions of the text that two or more agents independently flagged, detected by matching shared evidence spans, without being told in advance to look for agreement — and reduces the full set of findings to a deduplicated, stably-ordered list of **priority actions**, alongside an overall **risk band** (`low` / `moderate` / `elevated` / `critical`) derived from a fixed, disclosed scoring formula rather than a black-box aggregate judgment.

The same honesty invariants extend without exception: an empty submission is refused, not reviewed into a fabricated result; an agent with nothing classifiable in its slice of the input reports its own error and the overall response is marked incomplete rather than silently treated as clean; and every finding’s evidence remains a verbatim substring of the submitted input.

6 Discussion

The novelty this paper claims is specific and should not be overstated in either direction. It is not that deterministic pattern matching produces better or more insightful findings than a language model would — it plainly does not, and every lens’s own `attribution_note` says so. It is that a review framework carrying a named researcher’s identity has, for the first time in this project’s own experience building such tools, made it structurally impossible to produce a finding without also producing a machine-readable, type-enforced statement of whose methodology that finding actually represents. That is a claim about trust architecture and verifiability, not about analytical depth — and it is the specific gap that both classes of existing tooling described in §1 leave open.

Read against Papers 1 and 2, this paper completes a pattern across the series: each paper takes a construct that started as a named-but-unspecified theoretical instrument and documents its operationalization honestly, including the parts where the implementation’s own choices (a threshold, a keyword list, a scoring formula) are engineering judgment rather than a reading of the theory. Paper 2 did this for measurement; this paper does it for review.

7 Limitations

- **Keyword and pattern matching is not comprehension.** Every lens can miss a genuine concern phrased without its trigger vocabulary, and can false-positive on a trigger word used in an unrelated sense. This is disclosed per-lens via `attribution_note` rather than treated as a solved problem.

- **resonance is explicitly not CCET.** It reuses CCET’s general idea (compare passages, measure similarity) for a different question, with a cruder mechanism (shared words rather than shared meaning); conflating the two would overstate what the lens actually measures.
- **Provenance tagging discloses attribution, not correctness.** A finding tagged `laura-8layer-2025` is faithfully applying the primary author’s named categories; the tag says nothing about whether the underlying keyword heuristic correctly classified the passage in front of it.
- **The fifteen-agent layer’s risk-band scoring is a fixed formula, not a validated risk model.** It is disclosed as deterministic and reproducible, not as a calibrated estimate of real-world risk severity.
- **No external validation of review quality has been attempted.** As with Papers 1 and 2, everything reported here describes this framework’s own internal consistency and reproducibility, not an independent study of how its findings compare to expert human review.

8 Future Work

The most direct next step mirrors Paper 2’s own: an external, rerunnable benchmark comparing this framework’s findings against expert human review on a shared corpus of documents, so that “how good are these findings” becomes a question with an answer other than this project’s own confidence. A second, narrower step specific to this framework is extending the mandatory-attribution mechanism itself to any future lens drawing on a *different* named researcher’s published work, so the same three-way provenance discipline generalizes beyond this one framework’s current single source of theoretical grounding.

9 Conclusion

`call-laura` operationalizes the 8-Layer Model and the User Integrity Protocol from Paper 1 as a small, fully deterministic team of review agents, each a pure function of the submitted text, each anchoring every finding to a verbatim evidence span, and each required — at the type level, not by convention — to disclose whether its logic is the primary author’s own published framework, this project’s operationalization of a concept she names, or an addition unconnected to her work. Its fifteen-agent extension carries the identical discipline across a much broader set of practical review concerns, synthesizing convergent findings into one prioritized, honestly-partial review rather than fifteen disconnected verdicts. As with Papers 1 and 2, the claim made here is deliberately narrow: not that this is the most insightful reviewer available, but that it may be the most honest one available about exactly whose judgment produced each thing it says.

References

Reader is referred to Paper 1 (Serna Gaviria & Kepp, 2026) for the full theoretical bibliography, and to Paper 2 (Serna Gaviria & Kepp, 2026) for the Continuous Co-Evolution Tracker referenced in §4.1’s discussion of the **resonance** lens.